

Mobility project

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Introduction to the problem

The increasing availability of mobile phone data has enabled researchers and policymakers to analyze human mobility patterns at an unprecedented scale. Understanding population movements within a region is essential for various applications, including urban planning, transportation optimization, emergency response, and business decision-making. For Italy, datasets with this type of information are available, such as those provided by TIM Enterpriseⁱ.

In this project, we analyze a dataset that records the movements of individuals between different census areas (ACE) in Emilia-Romagna, Italy, based on mobile phone signals. The dataset provides demographic details such as age groups, gender, nationality, and the purpose of movement (business or consumer-related). Additionally, a separate dataset contains the geographical representation of these areas in GeoJSON format, allowing for spatial analysis and visualization.

The primary objectives of this study are twofold:

1. Data Visualization: Using geospatial visualization tools such as Folium and GeoPandas, we aim to create interactive maps that illustrate mobility patterns within the region.
2. Predictive Modeling: We develop forecasting models to predict the number of people moving between different areas, based on historical data. Such predictions can be useful for anticipating mobility trends and optimizing resource allocation.

Description of the state of the art

Typically, this type of dataset is used to analyze mobility within cities, for example, to understand which type of transportation is most commonly used by residents and in which areas of the city each type of transportation is predominantly usedⁱⁱ, or to estimate the direction of traffic within a cityⁱⁱⁱ.

In general, multiple datasets are used to extract different information depending on the type of analysis to be performed^{iv}. Additionally, by considering a larger scale, it is possible to predict people's migration flows^v.

Proposed approach

Below, we will describe the approach we undertook for forecasting the number of people moving in Emilia Romagna.

Data structure

671 CSV files were provided to us, containing all the movement data for the months of August and September 2019. In particular, the file with the fewest records is

`"fdestinationdata_20190925_000000_20190926_000000_08099022000000_08099027000000.csv"` containing 4.157 rows, and the file with the highest number of records is

`"fdestinationdata_20190927_000000_20190928_000000_08036046000001_08037016000000.csv"` containing 89.607 rows. The header of the files is layerid, datefrom, dateto, datavalue, toid, toname, Ni, Ns, Tb, Tc, Gm, Gf, F1, F2, F3, F4, F5 and F6.

Below is the explanation of the data contained in the files:

- Layerid and toid represent the starting geographical area and the destination geographical area, respectively, described by a combination of codes (Region|Province|Municipality|Area|Other).
- The datefrom represents the departure date grouped by hour, while dateto indicates the arrival date.
- The toname is the name of both the starting and destination geographical areas.
- Datavalue refers to the number of people moving from the layerid area to the toid area.

Datavalue is composed of the following:

- F1, ..., F6: For each F, the number of people in a specific age range moving.
- Gf: The number of females moving from layerid to toid.
- Gm: The number of males moving from layerid to toid.
- Ns: The number of non-Italian people moving from layerid to toid.
- Ni: The number of Italian people moving from layerid to toid.
- Tb: The number of people moving from layerid to toid for work (business).
- Tc: The number of people moving from layerid to toid for leisure (consumer).

In addition, the csv file "layers" contains the geographical data related to the ACE IDs of Emilia Romagna. The file contains 506 rows, and the header consists of: ID, GEOJSON.

- ID: contains the ID of the geographical area, which matches the layerid and toid in the previously mentioned files.
- GEOJSON: contains data in a geojson structure organized in the following way:

```
{
  "type": "Feature",
  "geometry": {
    "type": "MultiPolygon",
    "coordinates": [[[[[...]]]]]
  },
  "properties": {},
  "id": "..."
}
```

Data Storage and Cleansing

To carry out the project, we used PostgreSQL, as it is particularly well-suited for managing large datasets. Given the volume of the provided data, PostgreSQL, relational database management system (RDBMS) allows for efficient storage and manipulation of the information.

We created two tables: geom_data [Figure 1] and movements [Figure 2]. These tables were specifically designed to store geographical data and movement information, respectively.

```
CREATE TABLE geom_data (
  layerid VARCHAR(18) PRIMARY KEY,
  geojson JSONB
);
```

FIGURE 1: QUERY TO CREATE THE GEOM_DATA TABLE.

```

CREATE TABLE movements(
  layerid VARCHAR(18),
  datefrom TIMESTAMP,
  dateto TIMESTAMP,
  datavalue INT,
  toid VARCHAR(18),
  toname VARCHAR(100),
  Ni INT,
  Ns INT,
  Tb INT,
  Tc INT,
  Gm INT,
  Gf INT,
  F1 INT,
  F2 INT,
  F3 INT,
  F4 INT,
  F5 INT,
  F6 INT,
  PRIMARY KEY (layerid, toid, datefrom, dateto),
  FOREIGN KEY (layerid) REFERENCES geom_data(layerid)
);

```

FIGURE 2: QUERY TO CREATE THE MOVEMENTS TABLE.

Before loading the data into the tables, we reviewed and cleaned it. We used two Python files, *data_cleansing1.py* and *data_cleansing2.py*, to remove incorrect records. The *clean_data* function is responsible for cleaning and loading the data from multiple CSV files into a database called *project_mobility*. For each record, it extracts the departure date (*datefrom*) and arrival date (*dateto*). If the departure date is later than the arrival date, the record is ignored and not loaded into the database. Otherwise, the travel duration is calculated. Additionally, a consistency check is performed to ensure that the sums of the various categories of people match the total value of *datavalue*. If these sums align with each other and their total matches the value present in *datavalue*, the record is considered valid and inserted into the database. However, if there is a discrepancy between the calculated sums and the *datavalue*, the record is ignored and not loaded into the database. Since when we check the data insertion, we convert them from strings to floats, the float function returns an error if it has nothing inside. Therefore, the check is implicit when we clean the data. Automatic check for primary keys. We did not remove the outliers because August and September are special periods of the year, characterized by summer vacations. During these months, movement flows can vary significantly. Therefore, it did not seem justified to eliminate very high values, as they are not outliers caused by errors. In these months, in fact, there are days when few people travel between provinces, perhaps because they are already on vacation, and others when a very high number of movements is recorded, when many people return from their holidays. Records of trips that do not take place within the Emilia-Romagna region have been removed by using this query: *DELETE FROM movements WHERE toname NOT LIKE 'E%'*. The file *layer.csv* was imported into the database directly into the previously created table using the pgAdmin platform.

Graphical Representation

A graphical representation of the data was created in the file *Folium_visualization.ipynb* using the Folium library^{vi} and it was chosen to represent the total number of people departing from each *layerid* [Figure 3] and from each *toid*, divided by feature, with colors scaled from the minimum to the maximum numerical value present. Each feature is saved as a layer on the map.

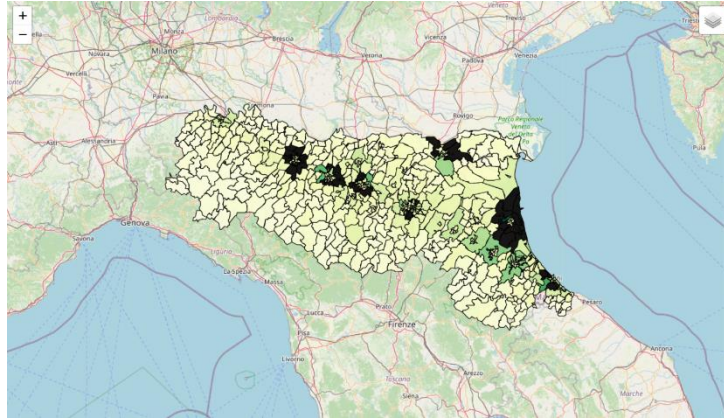


FIGURE 3: REPRESENTATION OF THE DATAVALUE FEATURE AT DEPARTURE.

Data analysis

Given the enormous amount of data in the database (229.622.575 records), a series of queries was selected to aggregate the information, making it more manageable for both analysis and forecasting. The first query (in *Regressions_layerid.ipynb*) considered represents the number of users leaving a certain ACE on a given date (returning 28.328 records).

The second query (in *Regressions_toid.ipynb*) represents the number of users entering a certain ACE on a given date (returning 28.548 records). The third query (in *Regressions_ACE.ipynb*) represents the number of users leaving one ACE to enter another ACE on a given date (returning 3.496.046 records). The final query (in *Regressions_province.ipynb*) represents the number of users leaving one province to enter another province on a given date (returning 4.941 records). In this paragraph, we present some of the analyses conducted; for a more detailed data analysis, please refer to the files *data_analysis_province.ipynb* and *data_analysis_ACE.ipynb*. For all the data subsets considered, a strong correlation was observed between the "datavalue" and the demographic data, as expected, given that the demographic data are fractions of the "datavalue" [Figure 4]. Most of the 'datavalue' values are concentrated near zero. The overall distribution of 'datavalue' is asymmetric, with a strong concentration of small values (less than 2.500) [Figure 5].

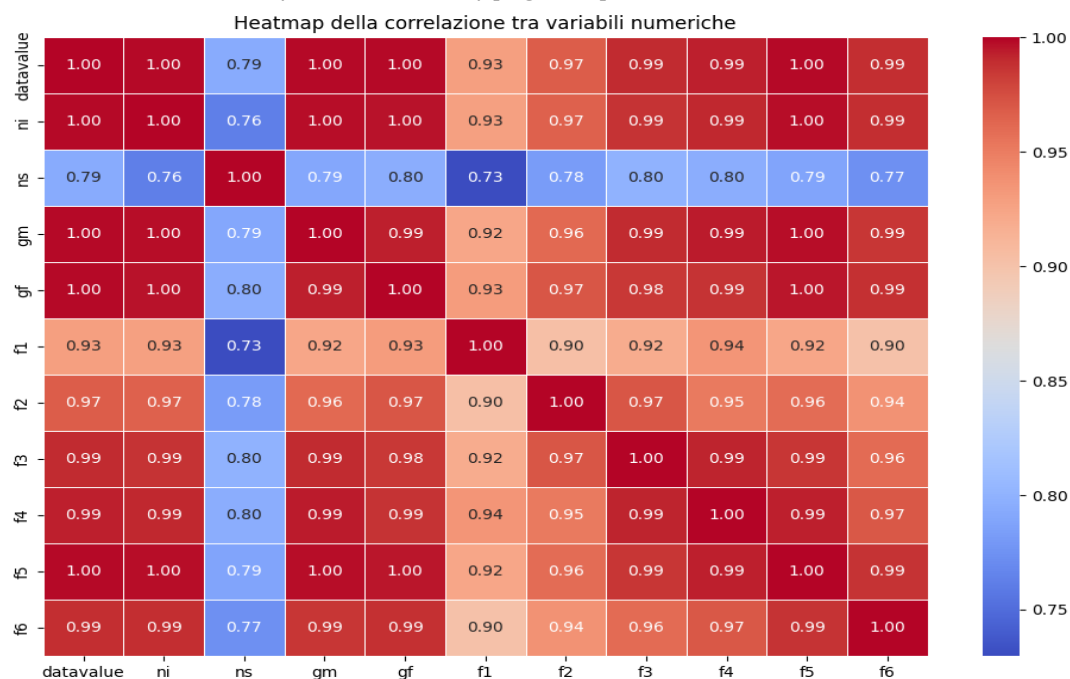


FIGURE 4: CORRELATION MATRIX, DATA_ANALYSIS_ACE.IPYNB

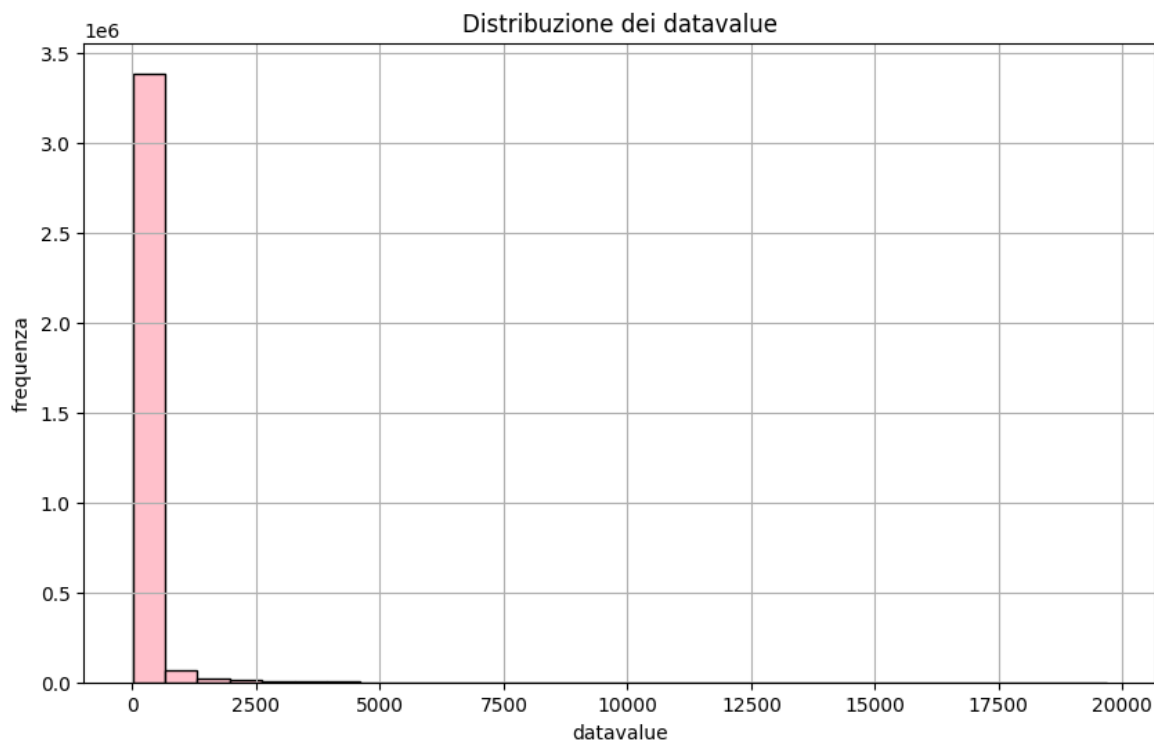


FIGURE 5: DATAVALUE DISTRIBUTION, DATA_ANALYSIS_ACE.IPYNB.

Forecasting models

We used three different regression models to predict datavalue, the number of people moving between different areas.

The feature tested were:

- Date: the date on each trip took place.
- Weekday: the day on each trip took place [0,6].
- Week: the week on each trip took place [0,9].
- Weekend: Whether the trip was on a weekend day.

The chosen models were Random Forest, K-Nearest Neighbors (KNN), and Gradient Boosting (GB)^{vii}. We chose to use these three regressions because they are particularly useful when working with nonlinear interactions between variables, as per our case.

The data was split into training and test sets with a 70% split for training and 30% for testing. We applied each model to each training data, made forecasting predictions on the test data, and evaluated the performance using the Mean Absolute Error (MAE) and the Mean Absolute Percentage Error (MAPE). Before applying the KNN model, we performed feature selection using SelectkBest^{viii} to choose the most relevant features. The data was then normalized using MinMaxScaler, and the target values were transformed using the logarithm, given that, from the graph, which allows visualizing the frequency of the datavalues, it is evident that their distribution follows an exponential trend. Where possible, to optimize the model parameters, we used GridSearchCV^{ix}, which determined the best parameters. The chosen parameters for KNN were: the number of neighbors ranging from two to fifty; weights: uniform, distance; metric: Euclidean, Manhattan, Minkowski. The chosen parameters for GB were a number of estimators (100, 300, 500), a learning rate (0.01, 0.1, 0.2), and a maximum depth (3, 5, 7). Regarding Random Forest, we decided not to use KSelector or automatic feature selection techniques because we preferred to manually handle the input features, based on domain knowledge and preliminary data analysis.

Additionally, we propose multi-output predictions that allow us to predict users moving from one province to another on a specific date based on gender, age group, nationality, and type of travel.

Results

For clarity sake, the results are presented in [Table 1](#).

TABLE 1: COMPUTATIONAL RESULTS

	Random forest		KNN k = 5		KNN k= 2		GB	
	MAE**	MAPE	MAE**	MAPE	MAE**	MAPE	MAE**	MAPE
Layerid	1027	11,21%	6551	124,39%	1632	16,20%	1003	10,52%
Toid	976	9,12%	6450	102,89%	1575	14,57%	929	8,56%
ACE	23	57,14 %	77	207,24%	31	48,68%	67	114,09%
Province*	11720	31,27%	4661	15,14%	/	/	5463	11,11%

Symbol legend:

* : Ran without select best with features date, profrom, pro to.

** : Rounded to the nearest integer.

/ : Does not apply.

- : couldn't be ran.

We chose to refer to the best performing MAPE value because MAPE penalises errors more on the low values of the target variable, which fits the distribution of datavalue.

The best performing regression across the board was Gradient Boosting.

Although MAE values are high, considering datavalue distribution, the results are considered acceptable in relation to MAPE values. Below, we show the graphs obtained using Gradient Boosting referring to layerid [[Figure 6](#)], toid [[Figure 7](#)], provinces [[Figure 8](#)]. Displayed are the predicted values y_{pred} and the values of y_{test} for the first 100 records.

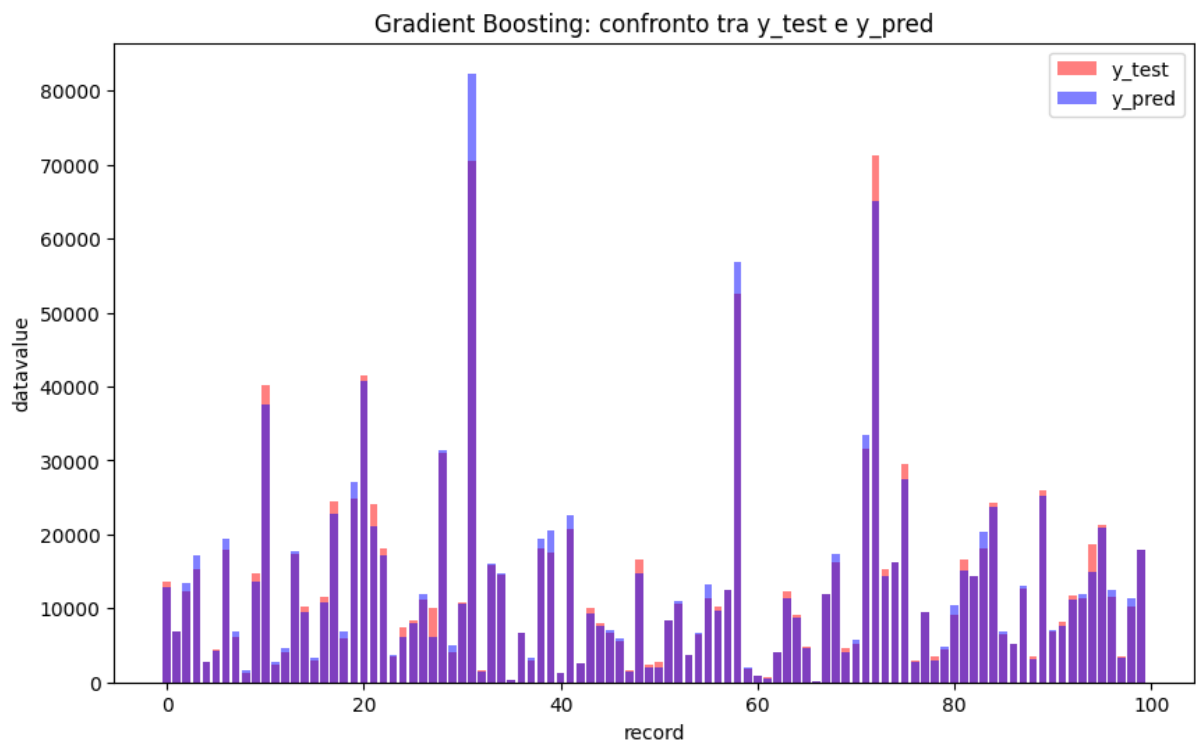


Figure 6: Comparison between y_test and y_pred obtained with Gradient Boosting, Regressions_layerid.ipynb.

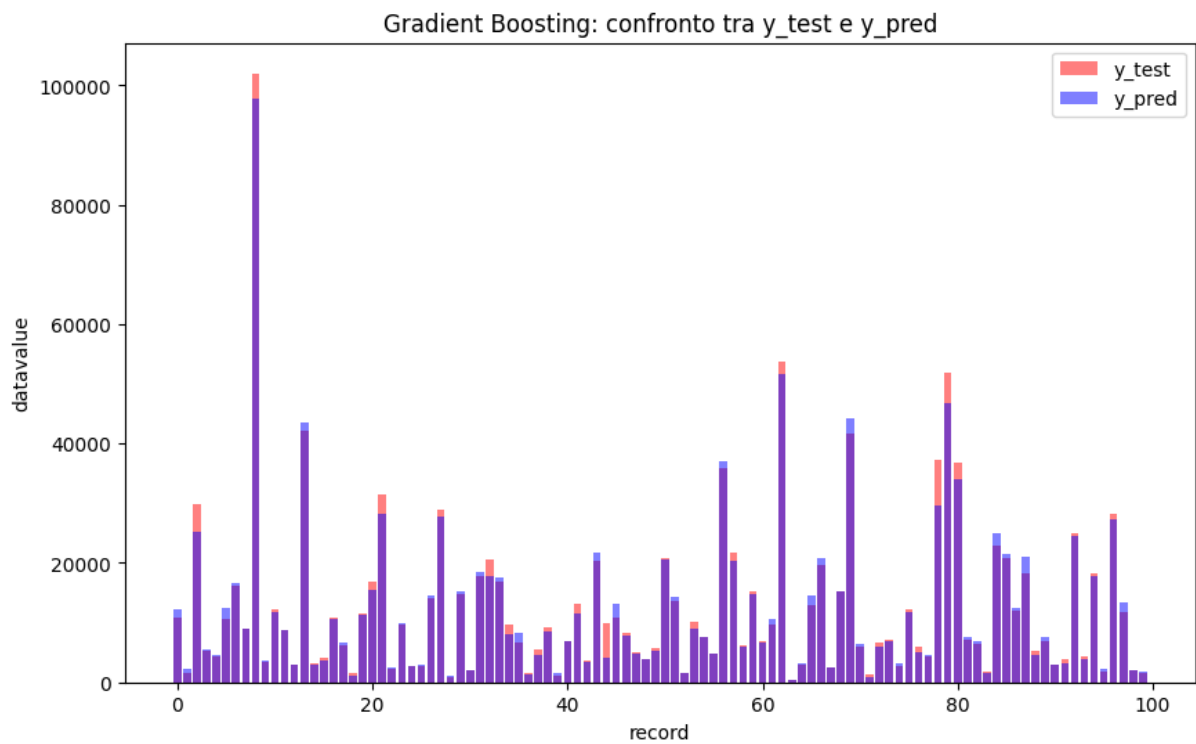


Figure 7: Comparison between y_test and y_pred obtained with Gradient Boosting, Regressions_toid.ipynb.

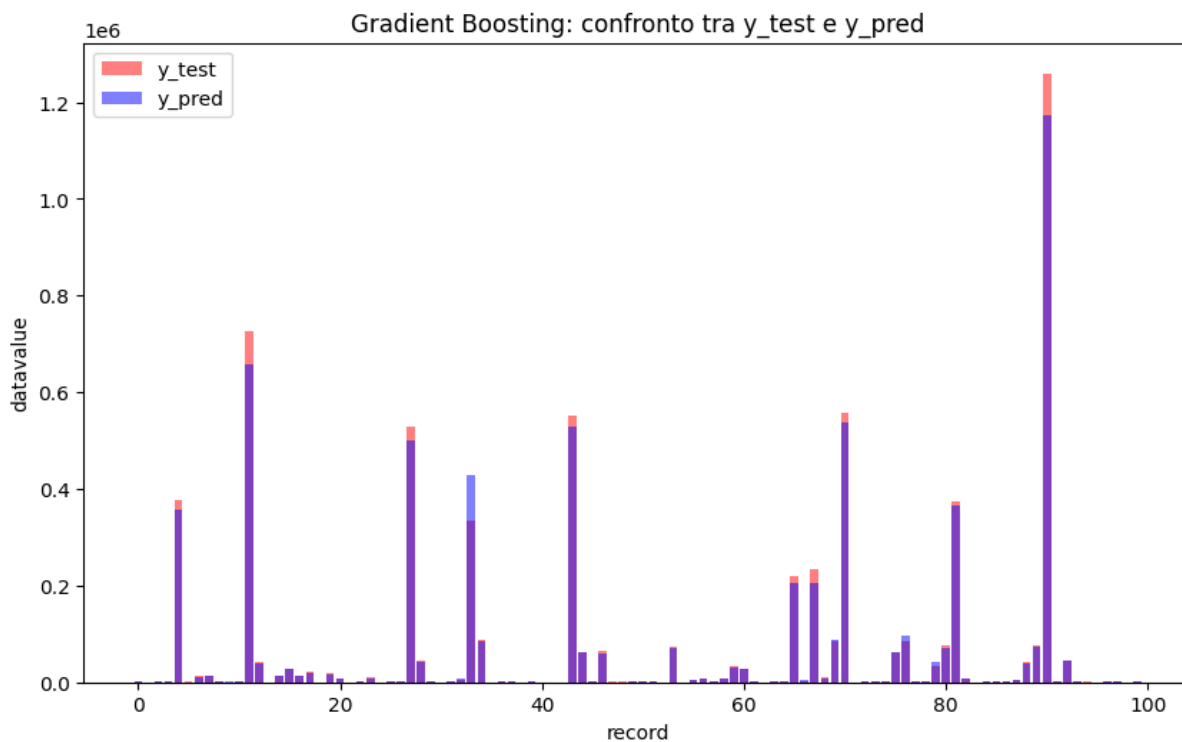


Figure 8: Comparison between y_{test} and y_{pred} obtained with Gradient Boosting, Regressions_province.ipynb.

In the table below [Table 2], we report the results of the regressions obtained with the multioutput models.

TABLE 2: RESULTS FOR MULTIOUTPUT REGRESSION

	Random Forest		KNN k=3		Gradient Boosting	
	MAE	MAPE	MAE	MAPE	MAE	MAPE
gf	5207	28,06%	2262	16,38%	4153	15,15%
gm	6753	25,27%	2837	15,23%	5793	14,01%
ni	14617	38,00%	4931	16,16%	8115	13,74%
ns	1128	/	333	/	804	/
tb	1805	/	909	/	1729	/
tc	8793	24,68%	4260	15,00%	9004	13,07%
f1	305	/	82	/	172	/
f2	1496	/	661	/	2132	/
f3	1634	/	699	/	1568	/

f4	2722	29,12%	1236	15,00%	2225	13,21%
f5	2972	36,35%	1257	14,89%	2257	13,48%
f6	4795	45,47%	1774	15,73%	3050	13,10%

Symbol legend:

/ : Not applicable due to values equal to 0 in y_{test} .

The results obtained from the multioutput models are not particularly satisfactory. One possible cause of this issue could be related to the fact that, for some categories, there are records with values equal to zero. When calculating the MAPE, the presence of zero values can cause a significant increase in the percentage error, as the MAPE formula involves dividing by the actual value (which, in these cases, is zero). This leads to very high errors, distorting the result and making it difficult to accurately assess the model's performance for these categories. The MAE alone provides an overview of the absolute error values, but it is not sufficient to fully evaluate the model's effectiveness.

ⁱ <https://www.timenterprise.it/5g-e-iot/data-analytics/tim-people-analytics>

ⁱⁱ Mizzi, C., Baroncini, A., Fabbri, A. et al. Individual mobility deep insight using mobile phones data. EPJ Data Sci. 12, 56 (2023). <https://doi.org/10.1140/epjds/s13688-023-00431-4>

ⁱⁱⁱ A. Scaloni and D. Micheli, "Estimation of mobility direction of a people flux by using a live 3G radio access network and smartphones in non-connected mode," 2015 IEEE 15th International Conference on Environment and Electrical Engineering (EEEIC), Rome, Italy, 2015, pp. 1869-1873, doi: 10.1109/EEEIC.2015.7165457. keywords: {Cities and towns;Delays;Roads;Mobile communication;Telecommunications;Data mining;Smart phones;smart city;flux of people;direction of mobility;cellular phone},

^{iv} Zhang, Desheng & Huang, Jun & Li, Ye & Zhang, Fan & Xu, Cheng-Zhong & He, Tian. (2014). Exploring human mobility with multi-source data at extremely large metropolitan scales. 10.1145/2639108.2639116.

^v [Migration_Forecasting_report.pdf](#)

^{vi} https://python-visualization.github.io/folium/latest/user_guide/geojson/geopandas_and_geo_interface.html

^{vii} [GradientBoostingRegressor — scikit-learn 1.6.1 documentation](#)

^{viii} [SelectKBest — scikit-learn 1.6.1 documentation](#)

^{ix} https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html

^x <https://www.diariodiunanalista.it/posts/cosa-e-la-grid-search-e-come-applicarla-in-python-con-sklearn/>